

Payment of 500\$	Time	Purchase on credit of 100\$	Time	Merchandise return of \$50
BEGIN TRANSACTION	t1		t1	
	t2	BEGIN TRANSACTION	t2	BEGIN TRANSACTION
Read BAL (500)	t3	Read BAL (500)	t3	
BAL = BAL - 500	t4	BAL = BAL + 100	t4	
	t5		t5	Read BAL (500)
Write BAL - COMMIT	t6		t6	BAL = BAL - 50
	t7		t7	
	t8	Write BAL - COMMIT	t8	
	t9		t9	Write BAL - COMMIT
	t10		t10	
	t11		t11	
	t12		t12	
	t13		t13	

Each of the three transactions read the customer record when the balance was \$500 (i.e., before any of the other transactions were completed and execute according to the timeline shown in the above figure.

- a. What balance will be included for the customer after the last transaction was completed?

\$450

- b. What balance should be included for the customer after the three transactions have been processed?

\$50

- c. Assume that only the first two transactions are executed and also assume that now the database enforces serializability using the two-phase locking mechanism. Illustrate what the process will be. Do any deadlocks occur?

Yes, there is a deadlock they both try to update the balance at the same time, so an exclusive lock is placed on both

- d. Now assume that only the first and the last transactions are executed and also assume again that the database enforces serializability using the two-phase locking mechanism. Illustrate what the process will be. Do any deadlocks occur?

No there is not a deadlock because of timing transaction 1 is finished and then t3 goes

- e. If versioning is used as concurrency control mechanism, what will happen? What will the final balance be?

The final balance will be \$50 because the first transaction will process and then the other two will fail because they are using an old version. Then transaction 2 and 3 will be reprocessed in order to allowing the balance to be made correct.